

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. Examination (EC)

May 2012

EC 308 Antenna & Wave Propagation

Date: 07.05.2012, Monday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Calculate the front to back ratio of an antenna in dB which radiates 3 kW in its most optimum direction and 500 watts in opposite direction 1
- (b) Calculate the approximate gain of parabolic reflector antenna operating at 4GHz, diameter 20 meters and illumination efficiency 55 %. 2
- (c) The radiation resistance of an antenna is 63Ω , loss resistance 7Ω . If antenna has power gain 16 dB then find directivity. 1
- (d) What three main factors determine the amount of refraction in the ionosphere? 1
- (e) What is the skip zone of a radio wave? 1
- (f) When a wide band of frequencies is transmitted simultaneously, each frequency will vary in the amount of fading. What is this variable fading called? 1
- Q - 2 (a) Explain principal of pattern multiplication. Using the pattern multiplication principle, explain the radiation pattern of 4 isotropic elements fed in phase, spaced λ 7
- (b) Design a broadside Dolph-Tschebyscheff array of 4 elements with spacing d between the elements and with a major to minor lobe ratio of 26 dB. Find the excitation coefficients and form the Array factor. 7

OR

- Q - 2 (a) Starting from retarded currents derive an expression for E and H components for a short dipole antenna if the spherical system is defined in r , θ and Φ . 10
- (b) Determine the distance from elementary dipole operating at 1Mhz at which radiation field is 6 times the induction field 4
- Q - 3 (a) In far field, for a small Loop Antenna derive field equations. Also derive Radiated Power and Radiation Resistance equation. 10
- (b) Sketch the helical geometry with its associated dimension showing relationship between circumference, spacing, turn length and pitch angle of helix 4

OR

- Q - 3 (a) Explain different layers of ionosphere 4
- (b) Prove that $R_r = 73 \Omega$ in $\lambda/2$ antenna 5

- (c) Derive the equation which gives the required shape of the Lens Antenna. 5

SECTION – II

- Q - 4 (a) The analysis model (microstrip patch antenna) that is the simplest is the: 1
(a) transmission line (b) cavity (c) integral equation/method of moments

- (b) In a microstrip antenna or line, the effective dielectric constant is used to take into account the: 1

(a) impedance of the radiating slots (b) fringing of the fields (c) geometry of the patch or center conductor of the line

- (c) Match the following 5

- | | |
|---------------------------|---------------------------|
| a) Friss formula | i) optimum pattern |
| b) Skelnoff theorem | ii) Circular polarization |
| c) Babinet's Principle | iii) Received Power |
| d) Tchebycheff Polynomial | iv) Null directions |
| e) Helical | v) Complimentary antennas |

- Q - 5 (a) Design a rectangular patch antenna at frequency 10 GHz using substrate with dielectric constant of 2.2, height of substrate $h=0.1588\text{cm}$. 8

- (b) Design 3 element Yagi Uda antenna at 20 GHz 3

- (c) Design Resonant antenna (leg length, tilt angle, height above ground) at frequency 22 GHz and β is 35° . 3

OR

- Q - 5 (a) Design log periodic antenna for FM broadcast band using ratio factor 0.95 and diameter equal to 0.08λ . 5

- (b) Explain Babinet's Principle. And discuss it for complementary antenna 4

- (c) Prove that for $\frac{\lambda}{2}$ folded dipole antenna the radiation resistance is approximate 280Ω (or 300Ω). Also prove $Z = 73n^2$ where Z is impedance of folded dipole antenna 5

- Q - 6 (a) Explain the slotted line carriage method (SWR) for impedance measurement of antenna under test 5

- (b) Describe various types of horn used at microwave frequency 7

- (c) Explain in brief duct propagation 2

OR

- Q - 6 (a) A parabolic antenna having circular mouth is to have gain of 1000 at $\lambda = 10\text{cm}$. Find the diameter of the mouth and half power beam width 5

- (b) Differentiate cassegrain feed and offset feed system in reflector antenna 4

- (c) Explain the experimental set ups for the measurement of gain of antenna by comparison 5